

Loop Invariants and the Correctness Proof

1 Loop Invariant ?

Loop invariant .

.

1. **Initialization**
loop invariant .
2. **Maintenance**
invariant , .
3. **Termination**
, invariant .

2 Insertion Sort Loop Invariant

Insertion Sort for ,
 $A[1 \dots j-1]$, .

2.1 Initialization

$j = 2$, $A[1 \dots j-1] = A[1]$.
, invariant .

2.2 Maintenance

$A[j]$
($A[j-1], A[j-2], \dots$) .
 $A[j]$ $A[1 \dots j]$.
 $A[1 \dots j-1]$.

2.3 Termination

for $j = n+1$.
invariant $A[1 \dots n]$.

, **Insertion Sort** .